

AMBA APB Design Specifications

August 2026

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Revision

Version	DoU	Descriptions
V1.0	July 10th, 2026	Final release AMBA APB Specifications
V1.1	July 18th, 2026	Update feature strobe mode, add input port PSTRB[3:0], and update function strobe mode for APB Specifications.
V1.2	July 20th, 2026	Update features Big Endian and Little Endian for APB Slave.

1. INTRODUCTION

1.1. About APB Protocol

The APB protocol is a low-cost interface, optimized for minimal power consumption and reduced interface complexity. The APB interface is not pipelined and is a simple, synchronous protocol. Every transfer takes at least two cycles to complete. The APB interface is designed for accessing the programmable control registers of peripheral devices. APB peripherals are typically connected to the main memory system using an APB bridge. For example, a bridge from AXI to APB could be used to connect several APB peripherals to an AXI memory system. APB transfers are initiated by an APB bridge. APB bridges can also be referred to as a Requester. A peripheral interface responds to requests. APB peripherals can also be referred to as a Completer. This specification will use Requester and Completer.

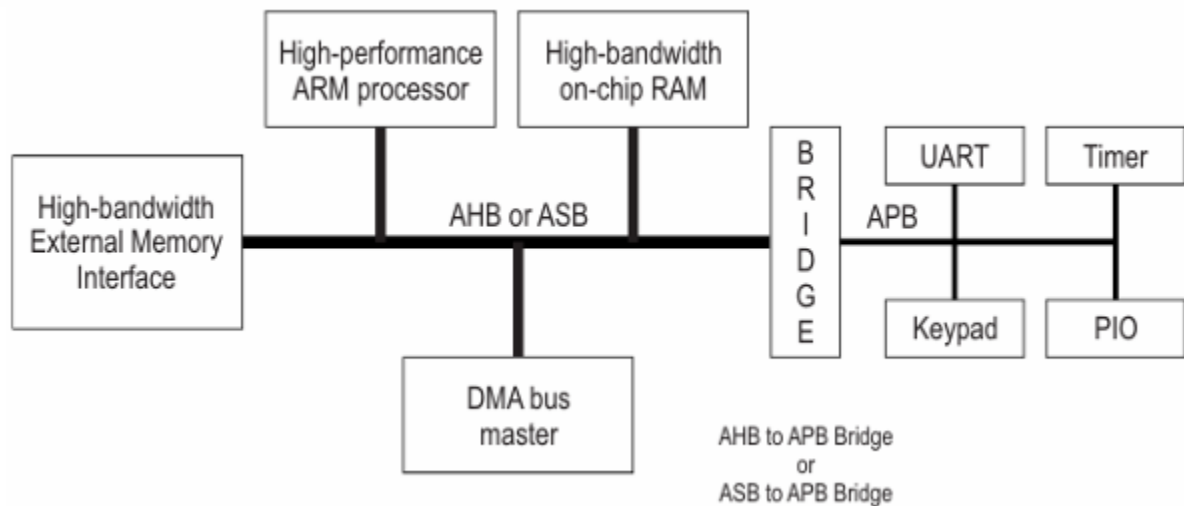


Figure 1.1. AMBA protocol

1.2. APB System Architecture

This section describes the APB system architecture.

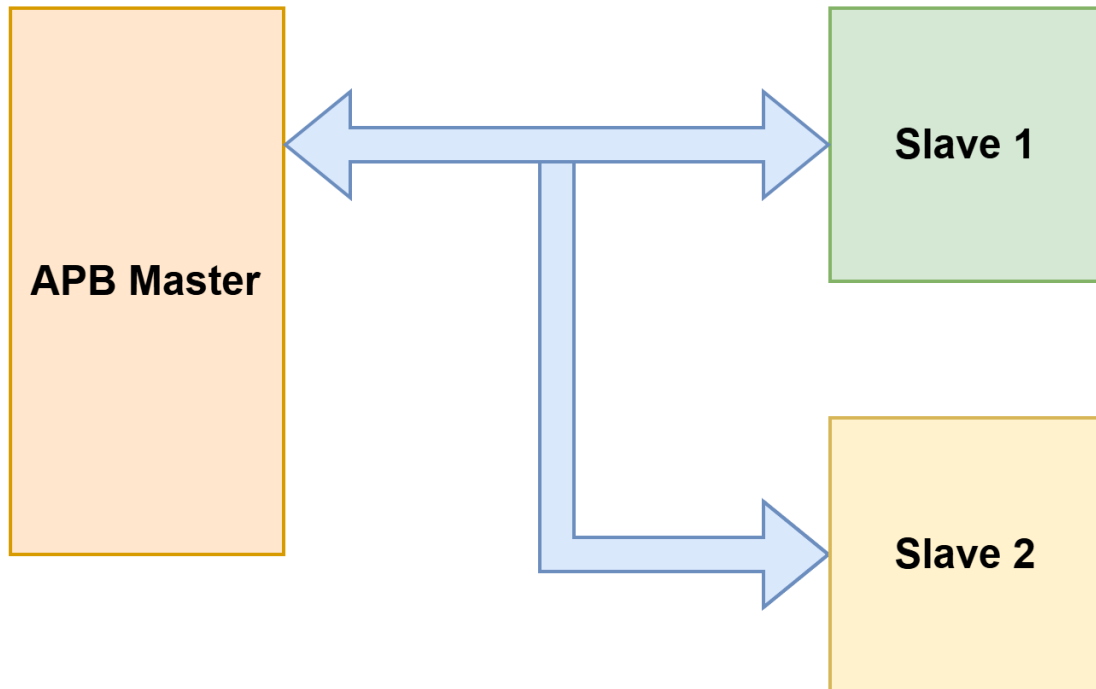


Figure 1.2. Design Architecture

The APB system consists of one APB Master and two APB Slaves. At any given transfer, the Master communicates with only one Slave, determined by the address configuration (see Section 2.4, Address Configuration).

For basic analysis, we will use ***Figure 1.3*** to examine the communication link between the APB Master and the APB Slave.

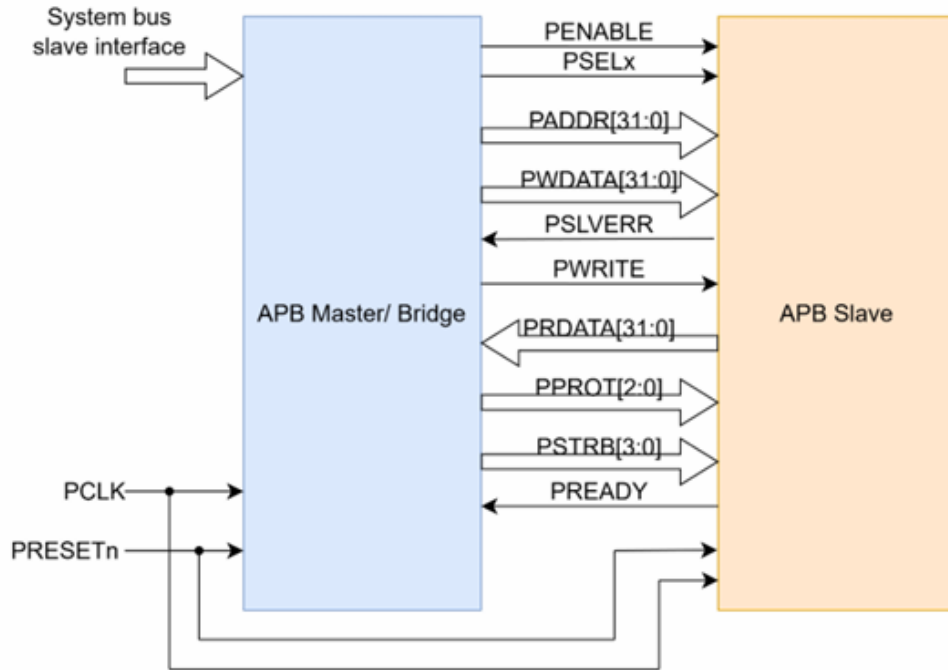


Figure 1.3. Block diagram and connection of APB Master and APB Slave

1.3. APB Interface Signals

This section describes the APB interface signals.

Some signals on the APB interface have a fixed width, and some can take a variety of widths. When the width is not fixed, it is described using a property. If the property value is zero, this means the signal is not present on the interface.

Table 1. Table of AMBA APB Signals

Signal	Source	Width	Description
PCLK	Clock Source	1	Clock. The rising edge of PCLK times all transfers on the APB
PRESETn	System Bus	1	Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal.
PADDR	APB bridge	ADDR_WIDTH	Address. This is the APB address bus. It can be up to 32 bits wide and is driven by the peripheral bus bridge unit.
PPROT	APB bridge	3	Protection type. This signal indicates the normal, privileged, or secure protection level of the transaction and whether the

			transaction is a data access or an instruction access.
PSELx	APB bridge	1	Slaves select signal; there will be one PSEL signal for each slave connected to master. It's an active high signal.
PENABLE	APB bridge	1	Enable. It indicates the 2nd cycle of data transfer. It's an active high signal.
PWRITE	APB bridge	1	Direction. PWRITE indicates an APB write access when HIGH and an APB read access when LOW.
PWDATA	APB bridge	DATA_WIDTH	Write data. Write data bus from Master to Slave, can be up to 32 bits wide
PSTRB	APB bridge	DATA_WIDTH/8	Write a strobe. PSTRB indicates which byte lanes to update during a write transfer. There is one write strobe for each 8 bits of the write data bus. PSTRB[n] corresponds to PWDATA [(8n + 7):(8n)]. PSTRB must not be active during a read transfer.
PREADY	Slave Interface	1	It is used by the slave to include wait states in the transfer. i.e. whenever slave is not ready to complete the transaction, it will request the master for some time by de-asserting the PREADY.
PSLVERR	Slave Interface	1	Indicates the Success or failure of the transfer. HIGH indicates failure, and LOW indicates Success.
PRDATA	Slave Interface	DATA_WIDTH	Read data us from Slave to Master, can be up to 32 bits wide

1.3.1. Address bus

An APB interface has a single address bus, PADDR, for read and write transfers. PADDR indicates a byte address.

PADDR is permitted to be unaligned with respect to the data width.

1.3.2. Data bus

The APB protocol has two independent data buses, PRDATA for read data and PWDATA for write data. The buses can be 8 bits, 16 bits, or 32 bits wide. The read and write data buses must have the same width.

Data transfers cannot occur concurrently because the read data and write data buses do not have their own individual handshake signals.

Note: In this design, the buses will be 32 bits wide.

2. Design Specifications

2.1. Design Features

This design will have the features below:

- Write/Read transfer.
- Wait state transfer.
- Multiple Slaves.
- Write strobes.
- Error response (PSLVERR).

2.2. High Level block design

This section describes the High-level block design of AMBA APB. Figure 2.1. Show the high-level block of AMBA APB, and the connection between APB Master and APB Slave.

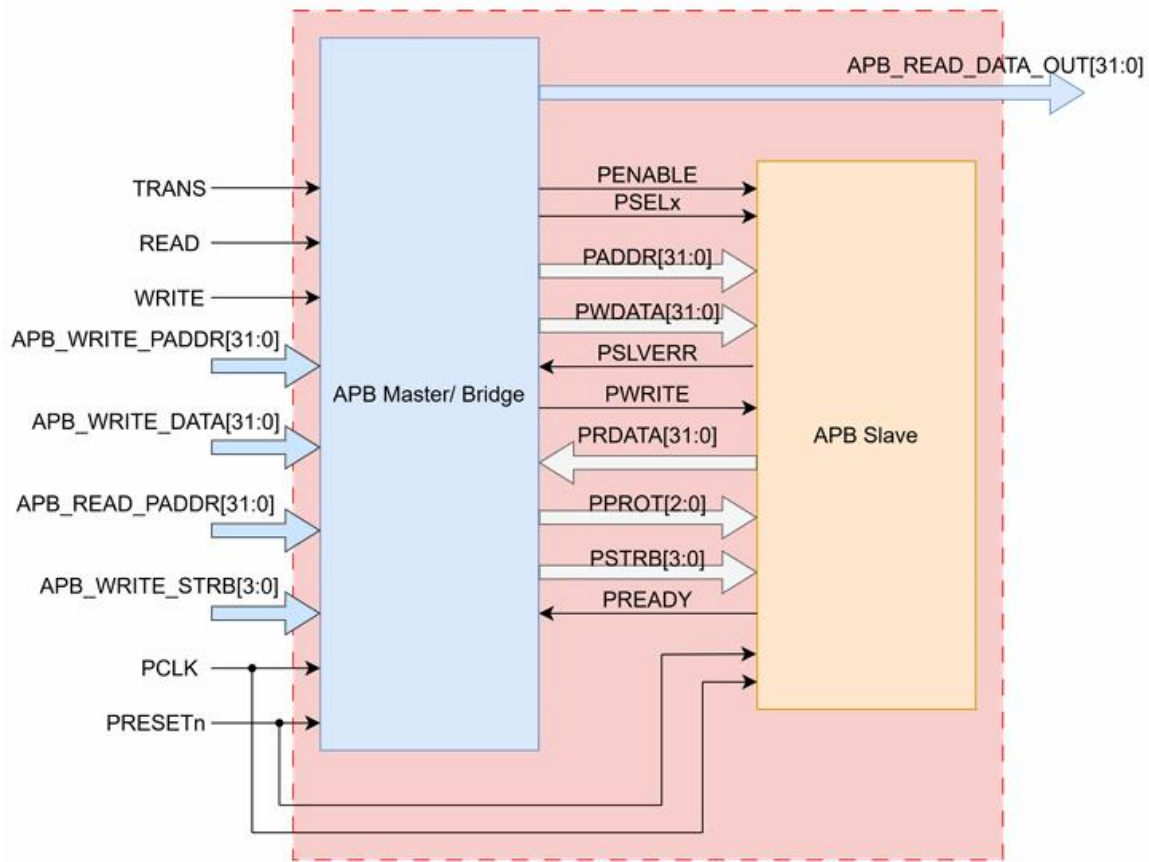


Figure 2.1. High level block design of AMBA APB

APB Master/Bridge: Acts as the controller, generating address, data, and control signals to communicate with the Slaves. Key signals include PADDR, PWDATA, PWRITE, PSELx, and PENABLE.

APB Slave: Represents the peripheral device that receives commands from the Master. It responds with read data (PRDATA), status signals (PREADY, PSLVERR), and executes read/write operations accordingly.

Control signals: Include PCLK (clock), PRESETn (reset), PENABLE (transfer enable), and PSELx (slave select).

Data and address paths: PADDR[31:0] carries the address, PWDATA[31:0] carries write data, and PRDATA[31:0] carries read data.

Transfer cycle: In each transaction, the Master selects only one Slave via PSELx, ensuring simple and sequential communication.

Table 2. Table of AMBA APB Interface signals (Real design)

Signal	Source	Width	Description
PCLK	Clock Source	1	Clock. The rising edge of PCLK times all transfers on the APB
PRESETn	System Bus	1	Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal.
TRANS	System Bus	1	Start signal. When the TRANS signal is set to HIGH, the system will begin operation. This signal may remain active for one clock cycle or multiple cycles.
READ	System Bus	1	Read enable signal.
WRITE	System Bus	1	Write enable signal.
APB_WRITE_PADDR	System Bus	ADDR_WIDTH	Address. This is the APB write address used by the APB master to write data to the slave. The address can be up to 32 bits wide.
APB_WRITE_DATA	System Bus	DATA_WIDTH	Data Write. This is the data sent from the APB master to the APB slave during a write operation. The data width can be up to 32 bits.
APB_READ_PADDR	System Bus	ADDR_WIDTH	Address. This is the APB read address used by the APB master to read data from the slave. The address can be up to 32 bits wide.
APB_READ_DATA_OUT	System Bus	DATA_WIDTH	Data Out. This is the data which is read from slave. The data width can be up to 32 bits.
PADDR	APB bridge	ADDR_WIDTH	Address. This is the APB address bus. It can be up to 32 bits wide and is driven by the peripheral bus bridge unit.
PRROT	APB bridge	3	Protection type. This signal indicates the normal, privileged, or secure

			protection level of the transaction and whether the transaction is a data access or an instruction access.
PSELx	APB bridge	1	Slaves select signal; there will be one PSEL signal for each slave connected to master. It's an active high signal.
PENABLE	APB bridge	1	Enable. It indicates the 2nd cycle of data transfer. It's an active high signal.
PWRITE	APB bridge	1	Direction. PWRITE indicates an APB write access when HIGH and an APB read access when LOW.
PWDATA	APB bridge	DATA_WIDTH	Write data. Write data bus from Master to Slave, can be up to 32 bits wide
PSTRB	APB bridge	DATA_WIDTH/8	Write a strobe. PSTRB indicates which byte lanes to update during a write transfer. There is one write strobe for each 8 bits of the write data bus. PSTRB[n] corresponds to PWDATA [(8n + 7):(8n)]. PSTRB must not be active during a read transfer.
PREADY	Slave Interface	1	It is used by the slave to include wait states in the transfer. i.e. whenever slave is not ready to complete the transaction, it will request the master for some time by de-asserting the PREADY.
PSLVERR	Slave Interface	1	Indicates the Success or failure of the transfer. HIGH indicates failure, and LOW indicates Success.
PRDATA	Slave Interface	DATA_WIDTH	Read data us from Slave to Master, can be up to 32 bits wide

2.3. APB Master

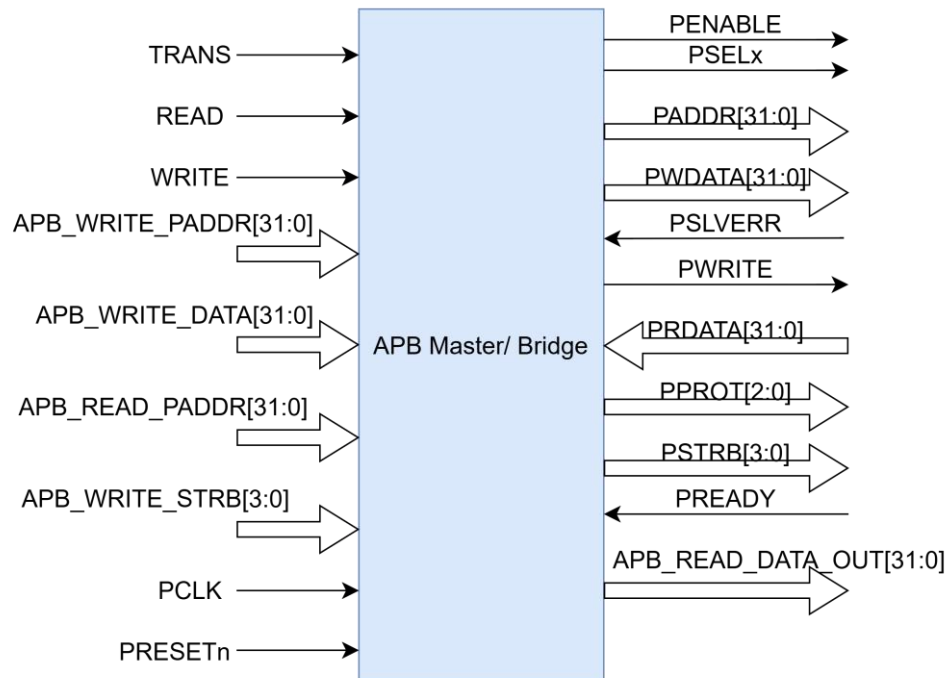


Figure 2.2. APB Master Interface

The **APB Master block** is responsible for generating the address, data, and control signals to communicate with the Slaves.

- **PADDR** is the bus carrying the address.
- **PWDATA** is the bus carrying write data from the Master to the Slave.
- **PRDATA** is the bus carrying read data from the Slave back to the Master.
- The control signal **PSELx** is used to select which Slave will communicate with the Master during a transfer.

In addition to these outputs, the APB Master receives several **input signals from the system** to initiate and manage transactions:

- **TRANS**: Indicates the start of a transfer.
- **READ**: Specifies a read operation request.
- **WRITE**: Specifies a write operation request.
- **APB_WRITE_PADDR[31:0]**: Provides the target address for a write transaction.
- **APB_WRITE_DATA[31:0]**: Provides the data payload for a write transaction.

- **APB_READ_PADDR[31:0]:** Provides the target address for a read transaction.
- **APB_WRITE_STRB[3:0]:** Byte strobe signals for write operations, indicating which bytes of the data bus are valid.
- **PCLK:** The clock input for synchronizing all APB operations.
- **PRESETn:** The active-low reset signal to initialize or reset the APB Master logic.

Together, these input signals drive the APB Master to generate the proper bus transactions, ensuring correct communication with the selected Slave.

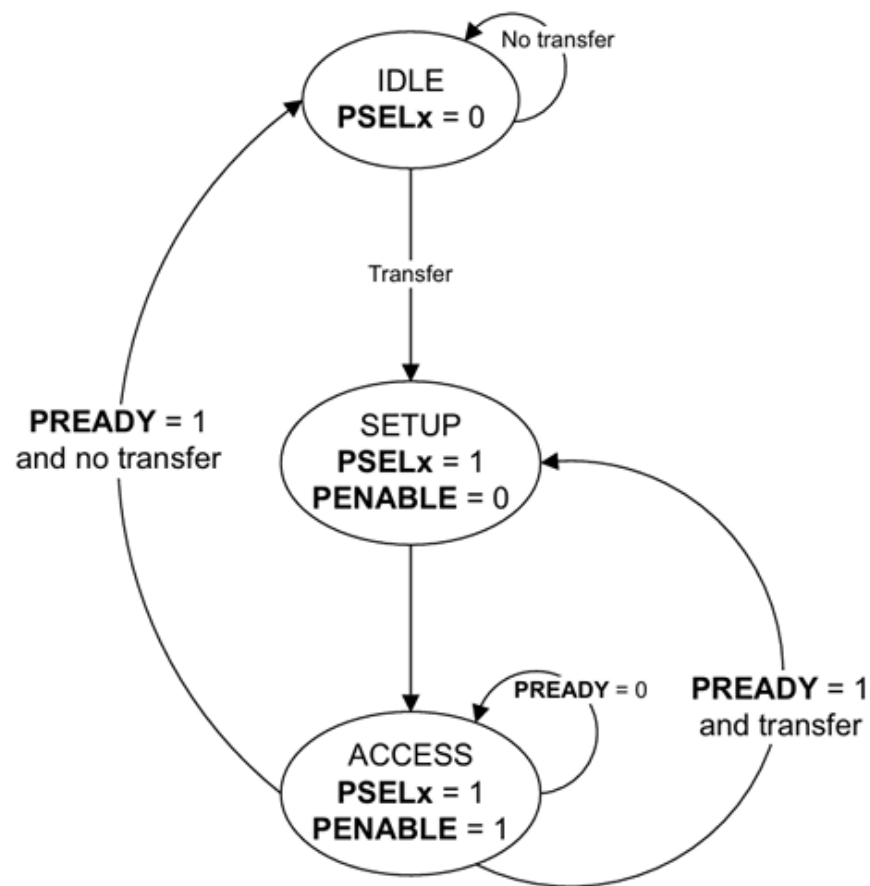


Figure 2.3. APB Master FSM diagram

IDLE: This is the default state of the APB interface.

SETUP: When a transfer is required, the interface moves into the SETUP state, where the appropriate select signal, PSELx, is asserted. The interface only remains in the SETUP state for one clock cycle and always moves to the ACCESS state on the next rising edge of the clock.

ACCESS: The enable signal, PENABLE, is asserted in the ACCESS state. The following signals must not change in the transition between SETUP and ACCESS and between cycles in the ACCESS state:

- PADDR
- PPROT
- PWRITE
- PWDATA, only for write transactions
- PSTRB

Exit from the ACCESS state is controlled by the PREADY signal from the Completer/Slave:

- If PREADY is held LOW by the Completer, then the interface remains in the ACCESS state.
- If PREADY is driven HIGH by the Completer, then the ACCESS state is exited and the bus returns to the IDLE state if no more transfers are required. Alternatively, the bus moves directly to the SETUP state if another transfer follows.

2.4. APB Slave

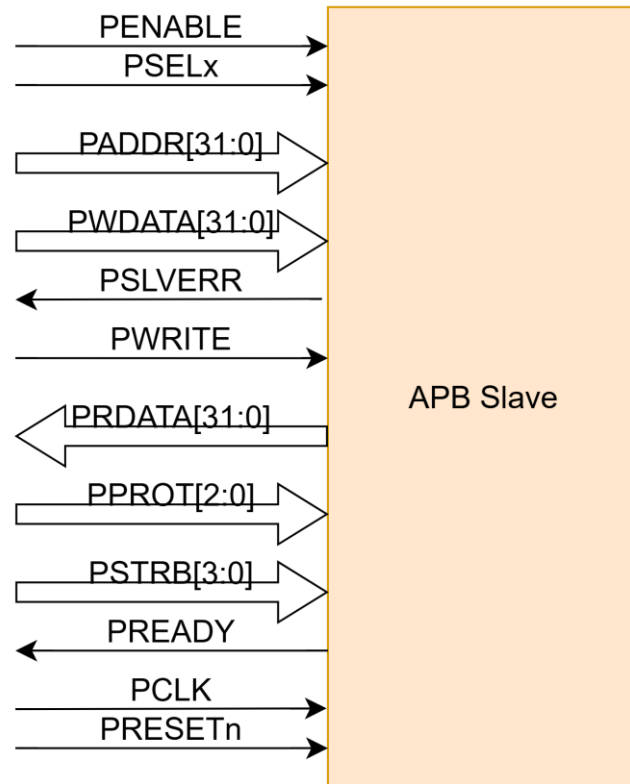


Figure 2.4. APB Slave Interface

The **APB Slave block** is responsible for receiving address, data, and control signals from the APB Master and executing the requested read or write operations. It also provides status and data back to the Master.

- **Input signals** from the Master/system:
 - **PCLK**: Clock input for synchronization.
 - **PRESETn**: Active-low reset to initialize the Slave logic.
 - **PSELx**: Slave select signal, indicating that this Slave is chosen by the Master.
 - **PENABLE**: Transfer enable signal, marking the active phase of a transaction.
 - **PADDR[31:0]**: Address bus specifying the target register or memory location inside the Slave.
 - **PWDATA[31:0]**: Write data bus carrying data from the Master to the Slave.
 - **PWRITE**: Control signal specifying whether the operation is a write (1) or read (0).

- PPROT[2:0]: Protection signals, often used for access control or privilege levels.
- PSTRB[3:0]: Byte strobe signals, indicating which bytes of the data bus are valid during a write.
- **Output signals** back to the Master:
 - PRDATA[31:0]: Read data bus, sending data from the Slave to the Master during read operations.
 - PREADY: Indicates whether the Slave is ready to complete the transfer.
 - PSLVERR

Address Configuration

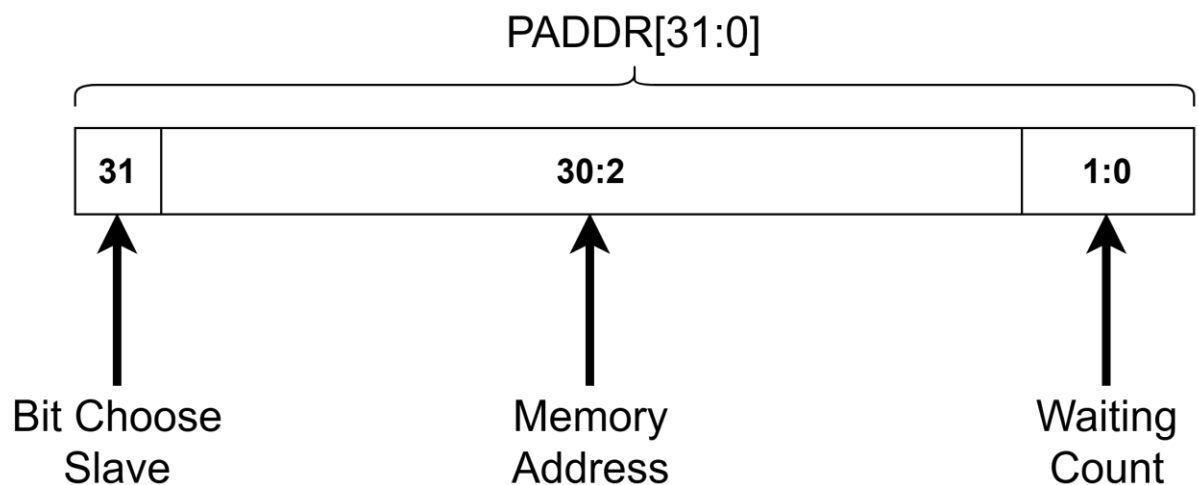


Figure 2.5. APB Address Configuration

Table 3. Table of Address Configuration

Bus Signals	Name	Descriptions
PADDR[31]	Slave Select Bit	Used to select the target slave. 0 selects Slave 0; 1 selects Slave 1.
PADDR[30:2]	Memory Address	These bits represent the internal memory index (Word-aligned). They are used to access specific data entries within the slave's memory array.
PADDR[1:0]	Waiting Count	These bits determine the number of wait states (delay cycles) the Slave will force the Master to wait before completing the transaction.

As shown in **Figure 2.5** and **Table 3**, the **APB address configuration** divides the 32-bit PADDR[31:0] signal into three functional fields:

- **PADDR[31] – Slave Select Bit:** Determines which Slave is targeted by the Master. A value of 0 selects Slave 0, while a value of 1 selects Slave 1.
- **PADDR[30:2] – Memory Address:** Represents the internal word-aligned memory index within the selected Slave. These bits are used to access specific registers or data entries.
- **PADDR[1:0] – Wait State Count:** Defines the number of wait cycles the Slave will insert before completing the transfer, allowing synchronization with slower peripherals.

This configuration ensures that each APB transaction is uniquely defined by the Slave selection, the target memory location, and the required wait states. It provides a simple yet flexible mechanism for addressing and timing control across multiple Slaves in the APB system.

In this design, the memory depth is **1024 words**, therefore the address field **PADDR[30:2]** must cover the range required to access all entries. The valid address range is:

- Minimum address: 32'h0.
- Maximum address: 32'h3FF.

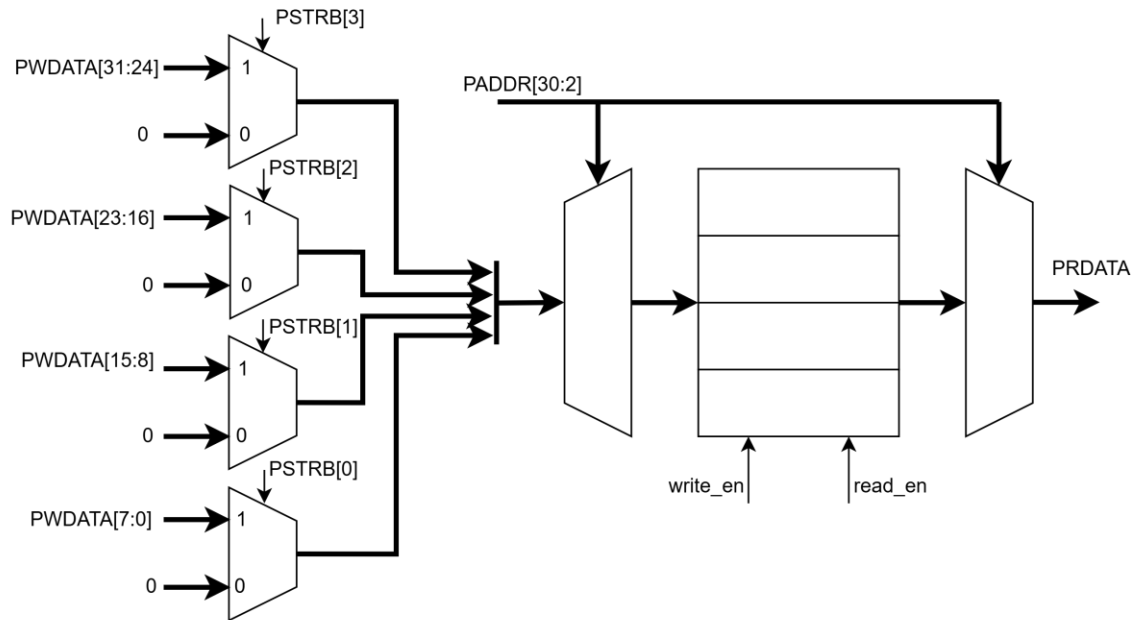


Figure 2.6. Memory Write and Read data diagram

The **APB Slave memory interface** implements byte-level write control and word-aligned addressing for read and write operations.

- **Write path:** Each byte of the 32-bit PWDATA bus is gated by a corresponding strobe signal PSTRB[3:0].
 - PSTRB[3] controls PWDATA[31:24]
 - PSTRB[2] controls PWDATA[23:16]
 - PSTRB[1] controls PWDATA[15:8]
 - PSTRB[0] controls PWDATA[7:0] When a strobe bit is asserted, the respective byte is passed to the memory block; otherwise, zero is written. This mechanism allows selective byte updates within a 32-bit word.
- **Memory block:** The memory array is addressed by PADDR[30:2], ensuring word alignment. Two control signals manage access:
 - write_en: Enables a write operation into the addressed location.
 - read_en: Enables a read operation from the addressed location.
- **Read path:** During a read transaction, the memory block outputs the 32-bit word at the specified address onto the PRDATA bus, which is then returned to the APB Master.

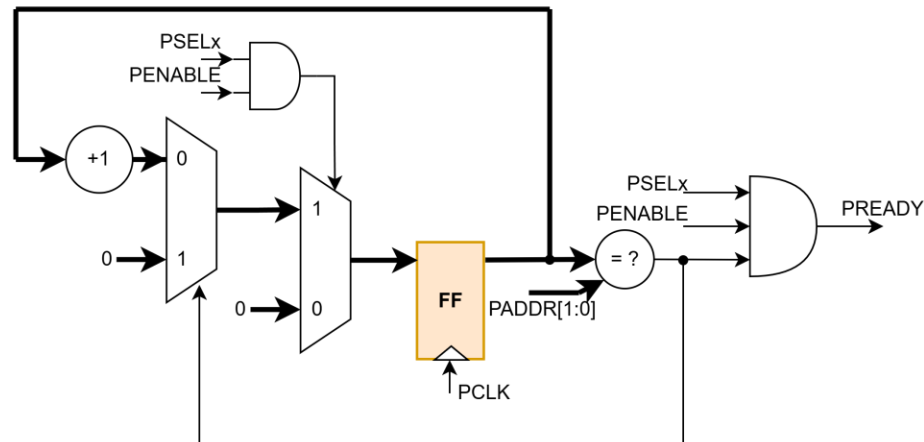


Figure 2.7. APB Slave Signal logic controller

This circuit is a **PREADY generation logic** block for the APB Slave.

Logic for PREADY:

PREADY is asserted (set high) when:

- The Slave is selected ($PSELx = 1$),
- The transfer is enabled ($PENABLE = 1$), and
- The counter value ($PADDR[1:0]$) matches the programmed wait count.

In other words, the Slave only signals readiness once the required number of wait cycles has elapsed, ensuring proper synchronization with the Master.

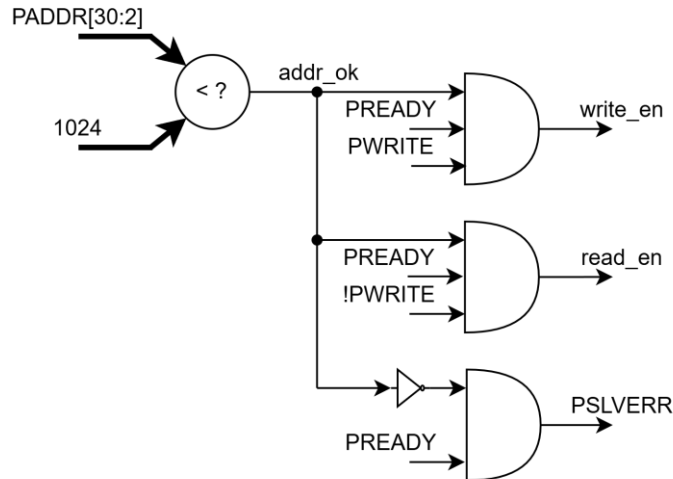


Figure 2.8. APB Slave Signal logic controller

This circuit is the **address to decode and access control logic** for the APB Slave.

The comparator checks whether $PADDR[30:2] < 1024$. If true, the signal `addr_ok` is asserted, meaning the address is valid within the memory depth.

When `addr_ok` is high and `PREADY` is asserted:

- If `PWRITE = 1`, the AND gate generates **write_en**, enabling a write operation.
- If `PWRITE = 0`, the AND gate generates **read_en**, enabling a read operation.
- If `addr_ok` is low (address out of range) but `PREADY` is high, the third AND gate asserts **PSLVERR**, signaling an error condition to the Master.

Logic for **PREADY** interaction:

- `PREADY` must be high for any operation to complete.
- Only when `addr_ok` is true does the Slave allow valid read or write operations.
- If `addr_ok` is false, **PSLVERR** is raised instead of performing the transaction.

3. System Operation

3.1. Basic Transfer (Write and Read transfer)

This section will describe basic transfer (write transfer and read transfer).

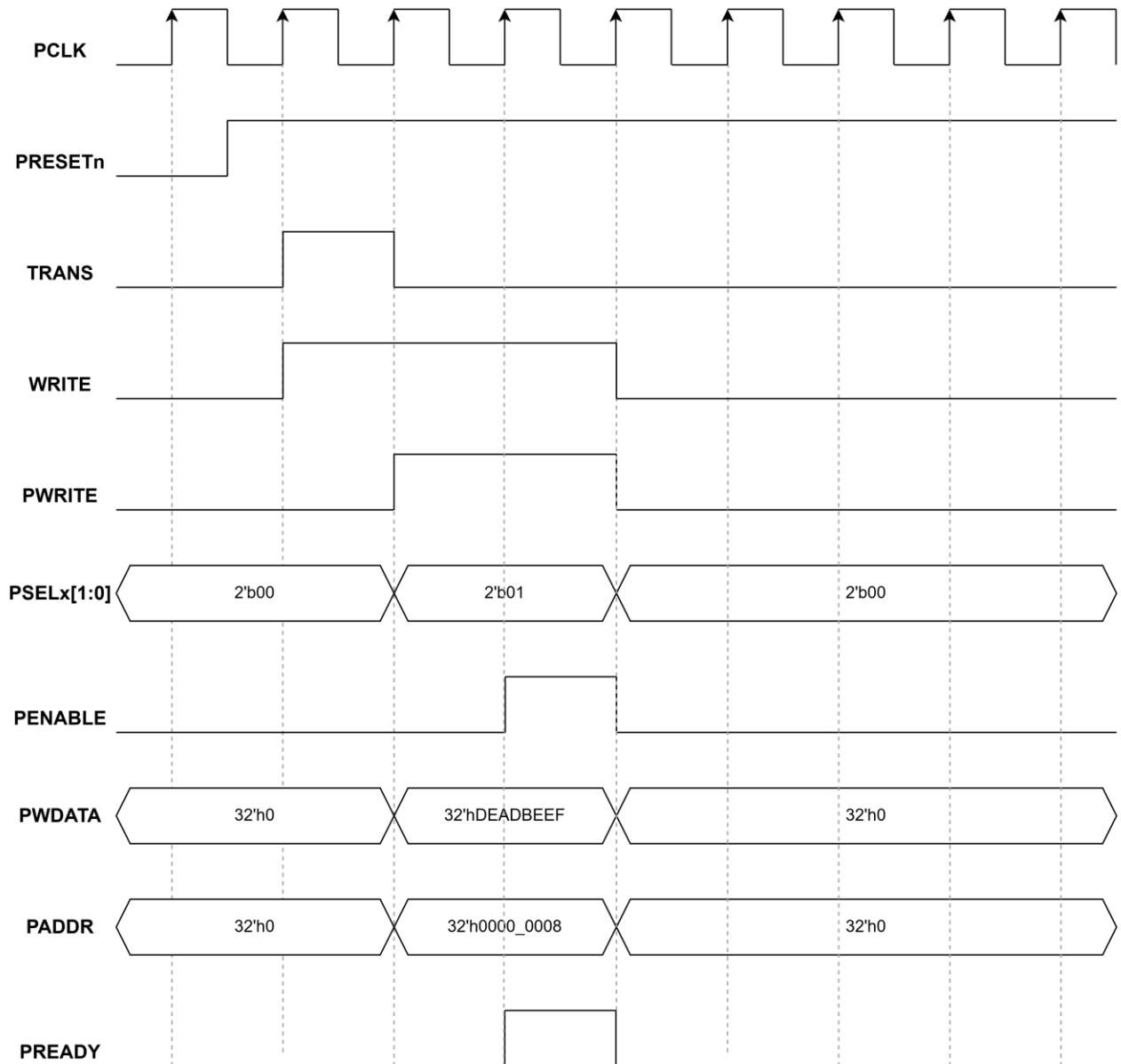


Figure 3.1. Write transfer with slave 1

Setup phase: The Master initiates the transaction by decoding the address. Based on PADDR[31], the appropriate Slave is selected (PSEL0 or PSEL1). At the same time, the Master places the address (PADDR) and, if it is a write transaction, the data (PWDATA) on the bus. During this phase, PSELx is active but PENABLE remains low.

Enable phase: In the following clock cycle, the Master asserts $PENABLE = 1$. At this point, the address and data are stable, and the selected Slave recognizes that the transaction has officially started.

Access phase: The Slave executes the requested operation. For a write, the data is stored at the specified address; for a read, the data is returned on the bus. When the Slave is ready, it asserts $PREADY = 1$ to indicate to the Master that the transaction is complete.

Completion: After the transaction finishes, the Master de-asserts both $PSELx$ and $PENABLE$. The address and data signals return to idle, and the bus becomes available for the next transaction.

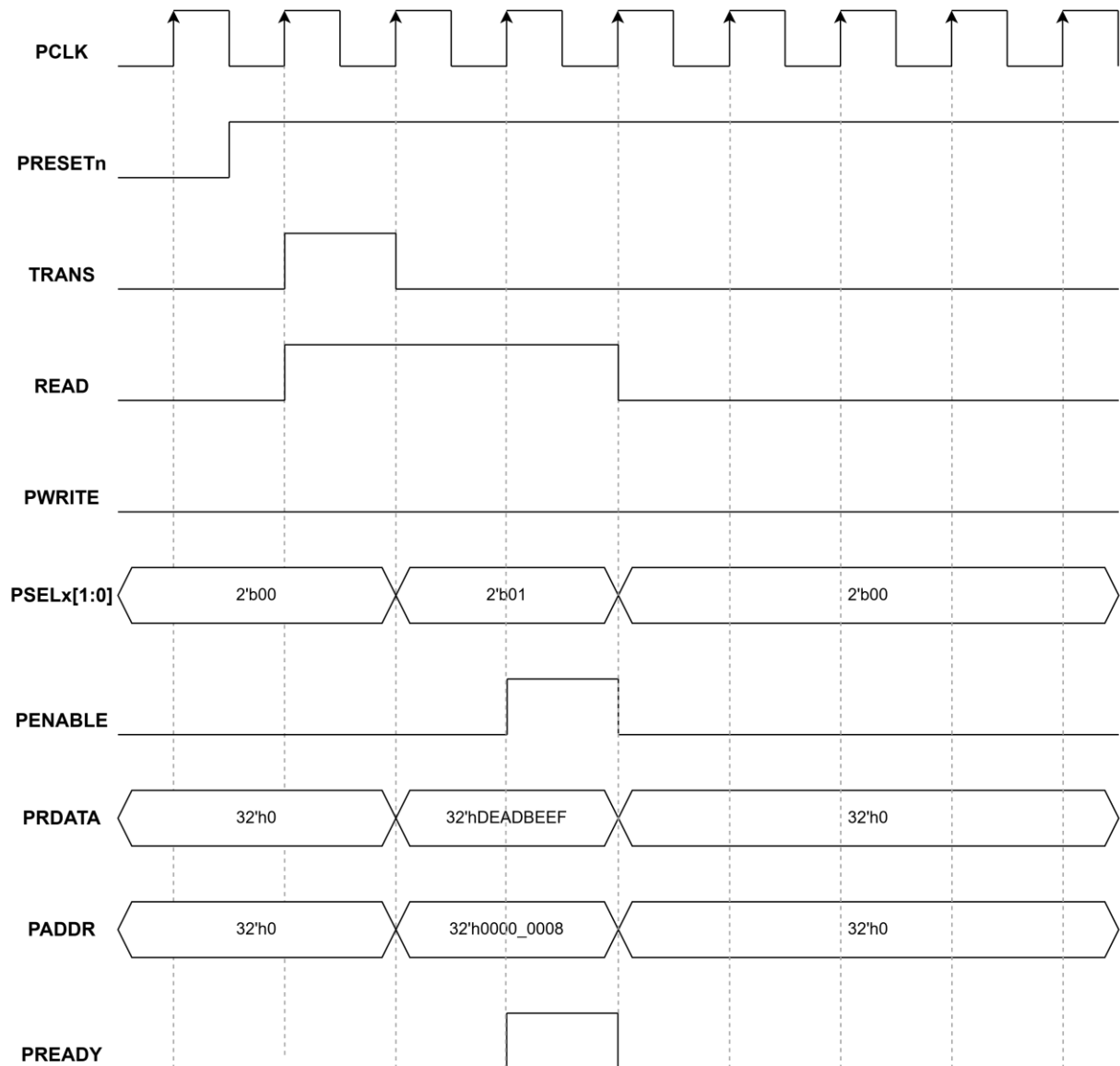


Figure 3.2. Read transfer with slave 1

Setup phase: The Master initiates the transaction by decoding the address. Based on PADDR[31], the appropriate Slave is selected (PSEL0 or PSEL1). At the same time, the Master places the target address (PADDR) on the bus and specifies the operation type (PWRITE = 0 for read, PWRITE = 1 for write). During this phase, PSELx is active but PENABLE remains low.

Enable phase: In the next clock cycle, the Master asserts PENABLE = 1. At this point, the address and data are stable, and the selected Slave recognizes that the transaction has officially started.

Access phase: The Slave executes the requested operation. For a write, the data is stored at the specified address; for a read, the data is returned on the bus. When the Slave is ready, it asserts $\text{PREADY} = 1$ to indicate to the Master that the transaction is complete.

Completion: After the transaction finishes, the Master de-asserts both PSELx and PENABLE . The address and data signals return to idle, and the bus becomes available for the next transaction.

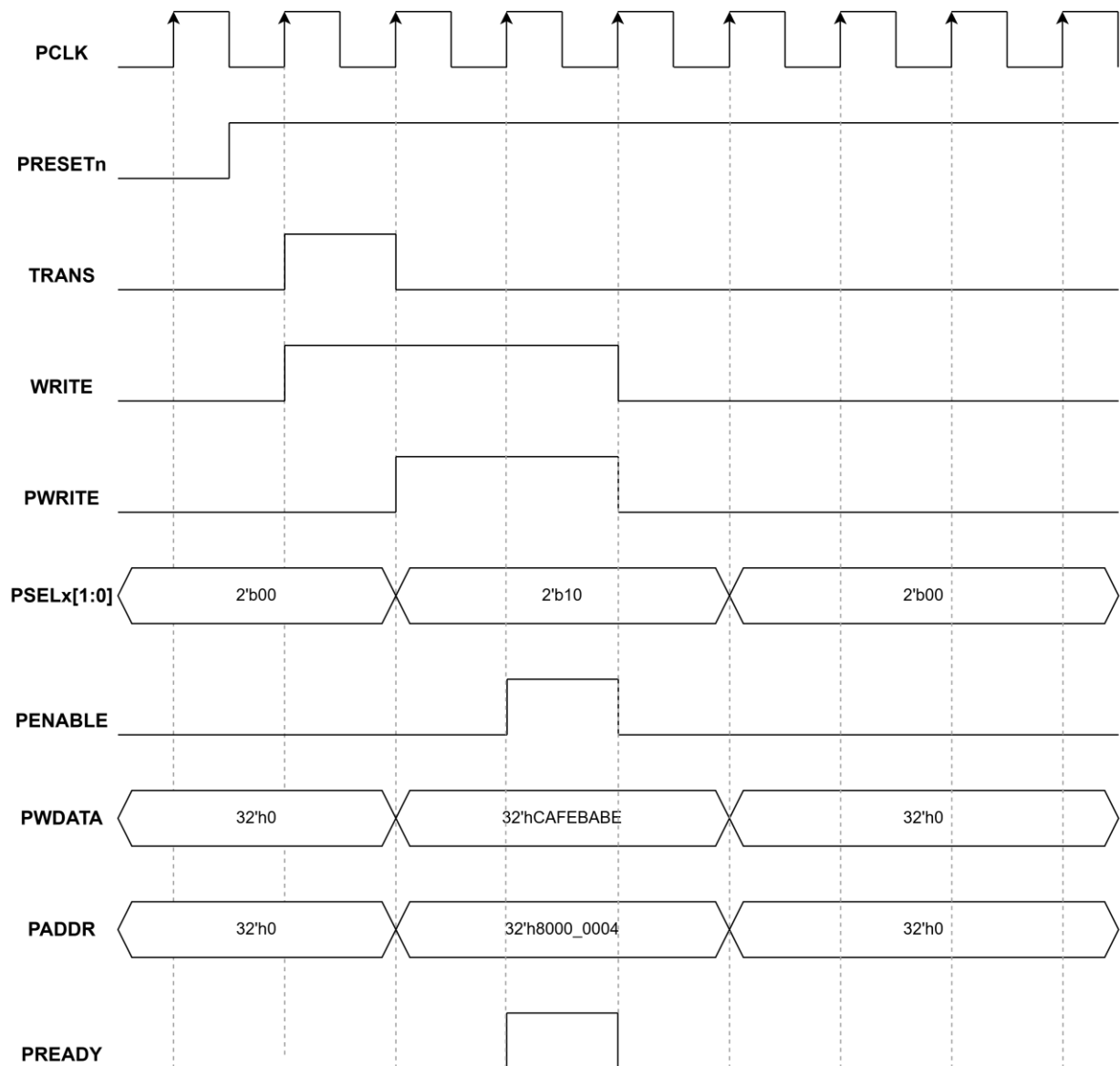


Figure 3.3. Write transfer with slave 2

Setup phase: The Master initiates the transaction by decoding the address. Based on PADDR[31], the appropriate Slave is selected (PSEL0 if bit 31 = 0, PSEL1 if bit 31 = 1). At the same time, the Master places the target address (PADDR) and the write data (PWRITE) on the bus, and sets PWRITE = 1 to indicate a write operation. During this phase, PSELx is active but PENABLE remains low.

Enable phase: In the next clock cycle, the Master asserts PENABLE = 1. At this point, the address and data are stable, and the selected Slave recognizes that the write transaction has officially started.

Access phase: The Slave processes the write request. When it is ready to accept the data, it asserts PREADY = 1. The data (for example, PWRITE = 0xCAFEBAFE) is then written into the specified address (PADDR = 0x8000_0004).

Completion: After the transaction is complete, the Master de-asserts both PSELx and PENABLE. The address and data signals return to idle, and the bus becomes available for the next transaction.

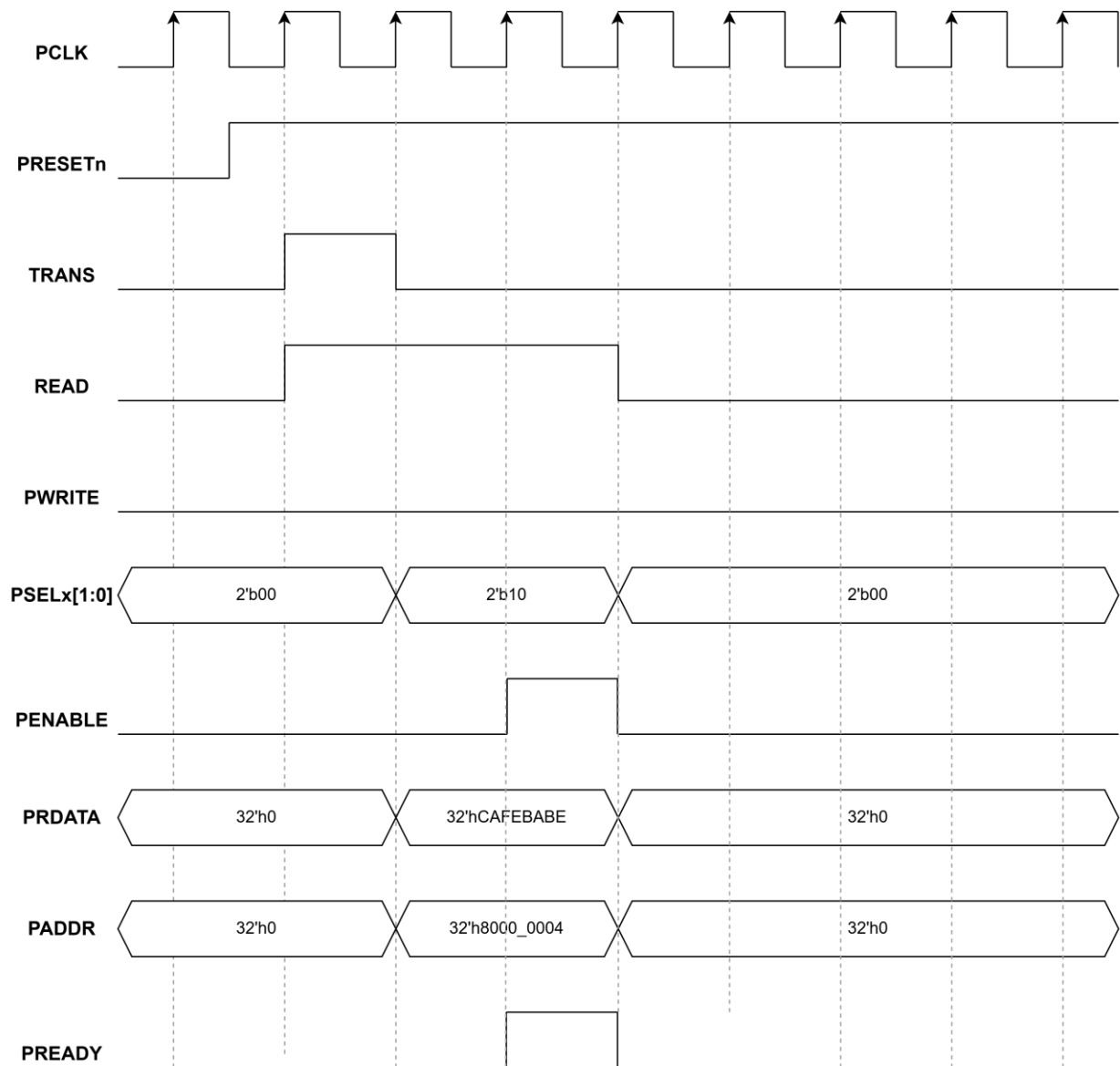


Figure 3.4. Read transfer with slave 2

Setup phase: The Master begins the transaction by decoding the address. Based on PADDR[31], the appropriate Slave is selected (PSEL0 if bit 31 = 0, PSEL1 if bit 31 = 1). At the same time, the Master places the read address (PADDR) on the bus and sets PWRITE = 0 to indicate a read. During this phase, PSELx is active but PENABLE remains low.

Enable phase: On the next rising edge of PCLK, the Master asserts PENABLE = 1. At this point, the address is stable, and the selected Slave recognizes that the read transaction has officially started.

Access phase: The Slave processes the read request. Since the peripheral is ready ($PREADY = 1$), it immediately drives the bus with the requested data (for example, $PRDATA = 0xCAFEBAFE$). The Master samples this data while $PREADY$ is high.

Completion: After the data is captured, the Master de-asserts both $PSELx$ and $PENABLE$. The address and data signals return to idle, and the bus is free for the next transaction.

3.2. Waiting Transfer (Write and Read transfer)

This section will describe the waiting state transfer for read and write transfer.

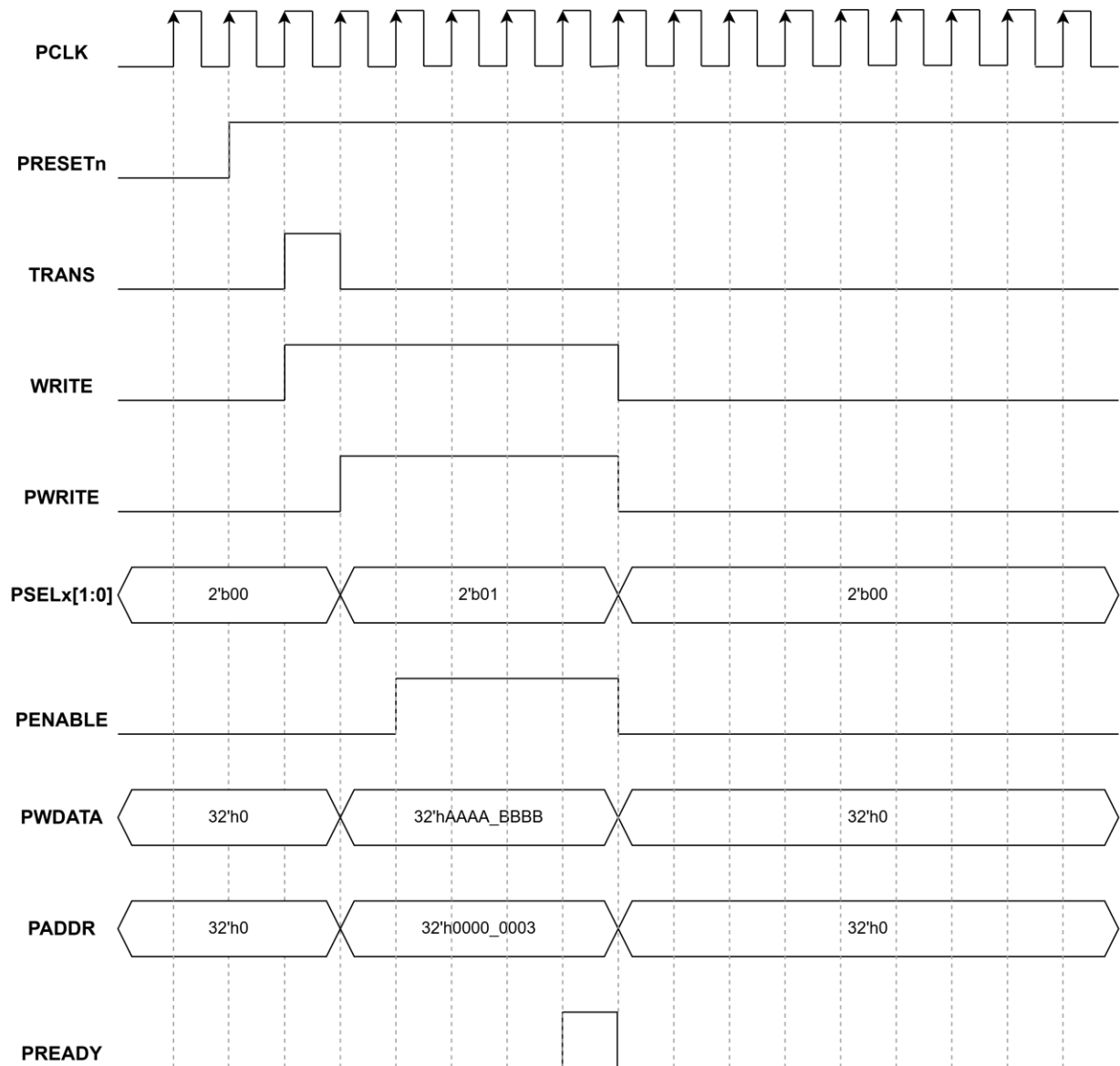


Figure 3.5. Write wait state transfer

Setup phase: The Master begins the transaction by decoding the address.

- **Slave selection:** The most significant bit of the address (PADDR[31]) determines which Slave is selected (PSEL0 if bit 31 = 0, PSEL1 if bit 31 = 1).

- **Wait state control:** The two least significant bits of the address (PADDR[1:0]) specify the number of wait cycles to be inserted before the Slave responds. At the same time, the Master places the target address (PADDR) and the write data (PWDATA) on the bus, and sets PWRITE = 1 to indicate a write operation. During this phase, PSELx is active but PENABLE remains low.

Enable phase: In the next clock cycle, the Master asserts PENABLE = 1. At this point, the address and data are stable, and the selected Slave recognizes that the write transaction has officially started.

Access phase: The Slave processes the write request. If wait states are required, the Slave keeps PREADY = 0 for the number of cycles indicated by PADDR[1:0]. When it is ready, the Slave asserts PREADY = 1. The data (for example, PWDATA = 0xAAAA_BBBB) is then written into the specified address (PADDR = 0x0000_0003).

Completion: After the transaction is complete, the Master de-asserts both PSELx and PENABLE. The address and data signals return to idle, and the bus becomes available for the next transaction.

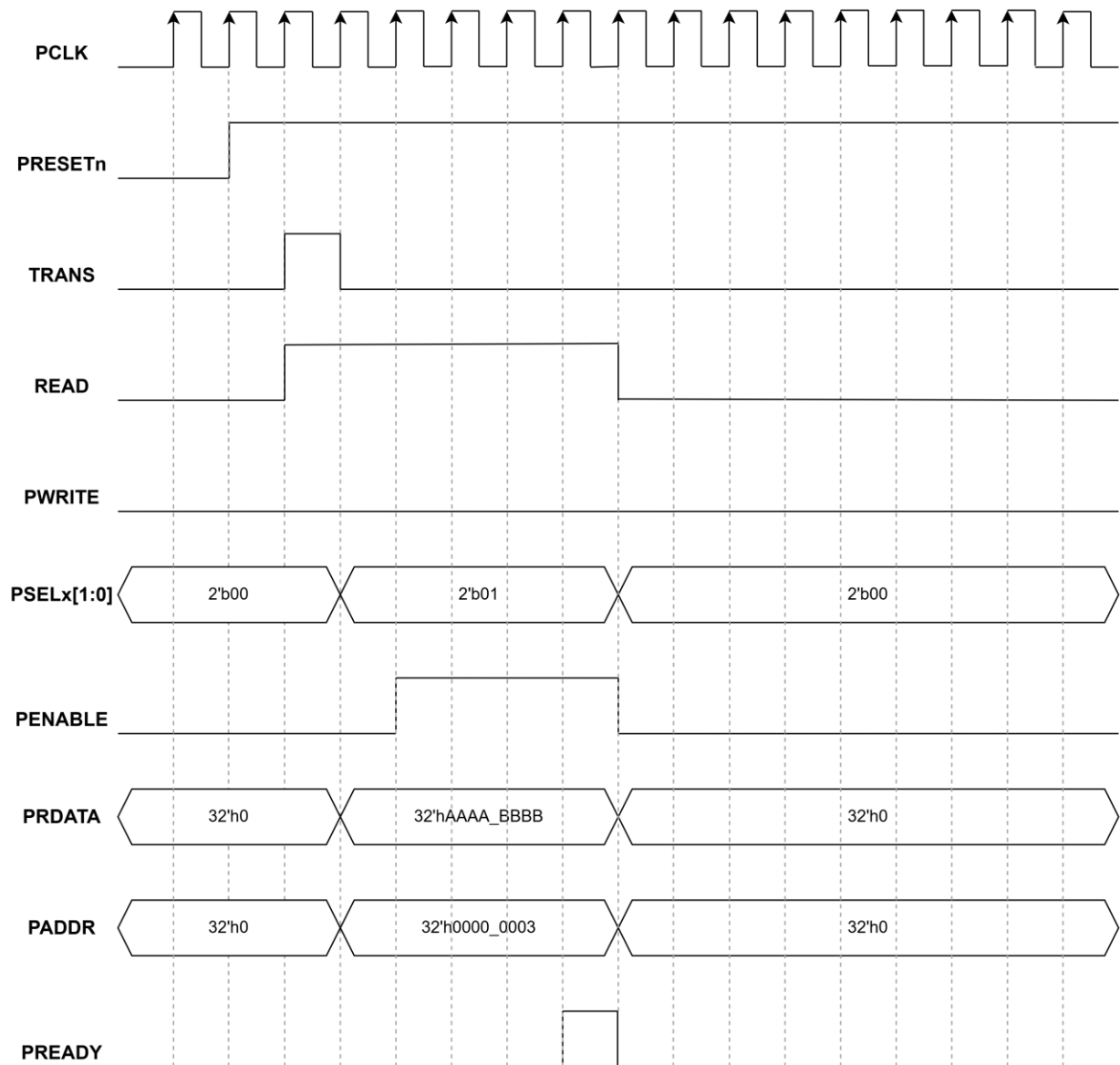


Figure 3.6. Read wait state transfer

Setup phase: The Master begins the transaction by decoding the address.

- **Slave selection:** The most significant bit of the address (PADDR[31]) determines which Slave is selected (PSEL0 if bit 31 = 0, PSEL1 if bit 31 = 1).
- **Wait state control:** The two least significant bits of the address (PADDR[1:0]) specify the number of wait cycles before the Slave responds. At the same time, the Master places the read address (PADDR) on the bus and

sets $PWRITE = 0$ to indicate a read operation. During this phase, $PSELx$ is active but $PENABLE$ remains low.

Enable phase: On the next rising edge of $PCLK$, the Master asserts $PENABLE = 1$. At this point, the address is stable, and the selected Slave recognizes that the read transaction has officially started.

Access phase: The Slave processes the read request. If wait states are required, it keeps $PREADY = 0$ for the number of cycles indicated by $PADDR[1:0]$. When ready, the Slave asserts $PREADY = 1$ and drives the bus with the requested data (for example, $PRDATA = 0xAAAA_BBBB$). The Master samples this data while $PREADY$ is high.

Completion: After the data is captured, the Master de-asserts both $PSELx$ and $PENABLE$. The address and data signals return to idle, and the bus is free for the next transaction.

3.3. APB Error Transfer (PSLVERR)

This section will describe the feature Error response of APB Slave (Error transfer).

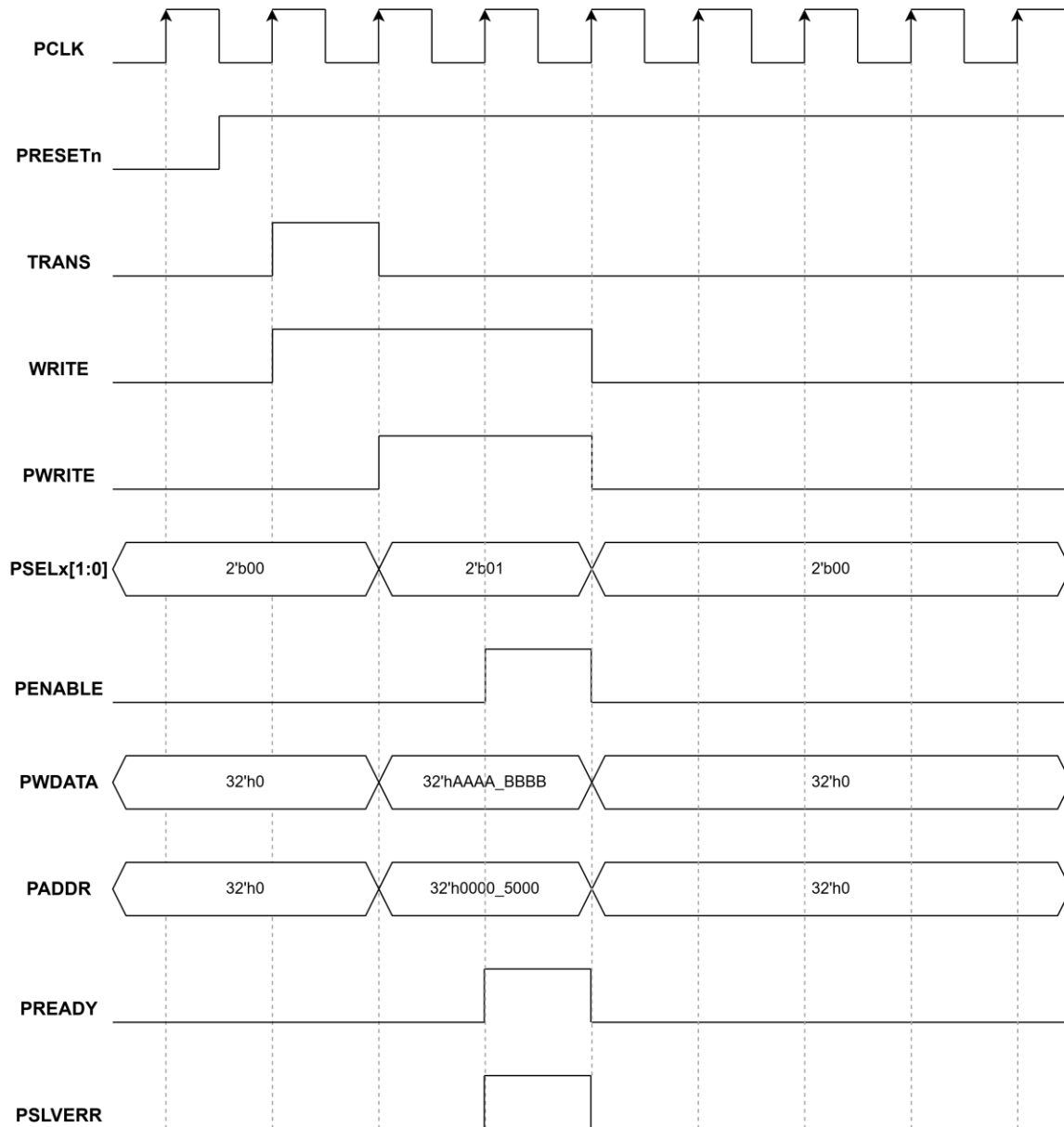


Figure 3.7. Error response with write transfer

Setup phase: The Master begins the transaction by decoding the address.

Slave selection: The most significant bit of the address (PADDR[31]) determines which Slave is selected (PSEL0 if bit 31 = 0, PSEL1 if bit 31 = 1).

Error detection: The middle field of the address (PADDR[30:2]) is checked against the valid range. If $\text{PADDR}[30:2] > 1024$, the address is invalid. At the same time, the Master places the target address (PADDR) on the bus and sets the operation type (PWRITE = 1 for write). During this phase, PSELx is active but PENABLE remains low.

Enable phase: On the next rising edge of PCLK, the Master asserts PENABLE = 1. At this point, the address is stable, and the selected Slave recognizes that the transaction has officially started.

Access phase: The Slave checks the validity of the address.

- If $\text{PADDR}[30:2] \leq 1024$, it proceeds normally.
- If $\text{PADDR}[30:2] > 1024$, the Slave asserts PSLVERR = 1 together with PREADY = 1 to indicate an error response. No valid data is transferred.

Completion: The Master samples PSLVERR along with PREADY. Since PSLVERR = 1, the transaction is considered failed. The Master then de-asserts both PSELx and PENABLE. All signals return to idle, and the bus is free for the next transaction.

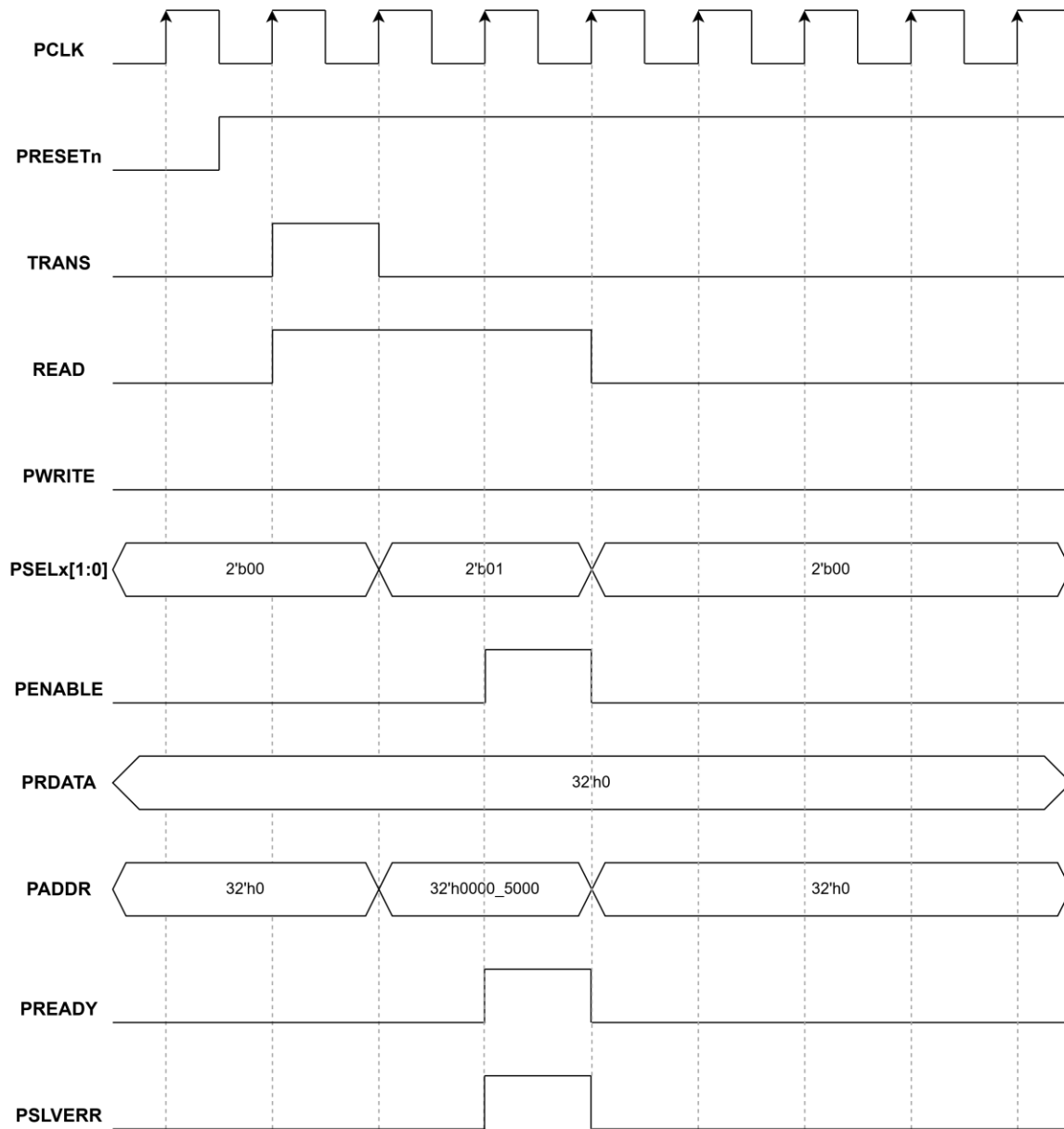


Figure 3.8. Error response with read transfer

Setup phase: The Master begins the transaction by decoding the address.

Slave selection: The most significant bit of the address (PADDR[31]) determines which Slave is selected (PSEL0 if bit 31 = 0, PSEL1 if bit 31 = 1).

Error detection: The middle field of the address (PADDR[30:2]) is checked against the valid range. If PADDR[30:2] > 1024, the address is invalid. At the same time, the Master places the target address (PADDR) on the bus and sets the

operation type ($PWRITE = 0$ for read). During this phase, $PSELx$ is active but $PENABLE$ remains low.

Enable phase: On the next rising edge of $PCLK$, the Master asserts $PENABLE = 1$. At this point, the address is stable, and the selected Slave recognizes that the transaction has officially started.

Access phase: The Slave checks the validity of the address.

- If $PADDR[30:2] \leq 1024$, it proceeds normally.
- If $PADDR[30:2] > 1024$, the Slave asserts $PSLVERR = 1$ together with $PREADY = 1$ to indicate an error response. No valid data is transferred.

Completion: The Master samples $PSLVERR$ along with $PREADY$. Since $PSLVERR = 1$, the transaction is considered failed. The Master then de-asserts both $PSELx$ and $PENABLE$. All signals return to idle, and the bus is free for the next transaction.